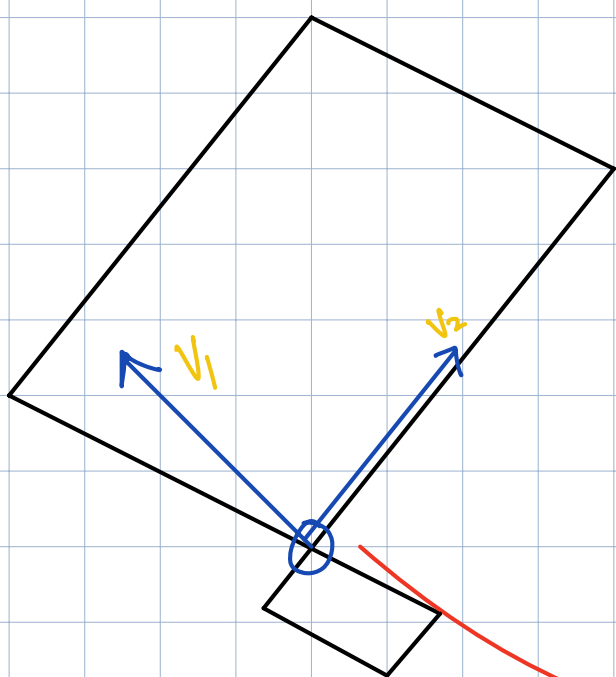
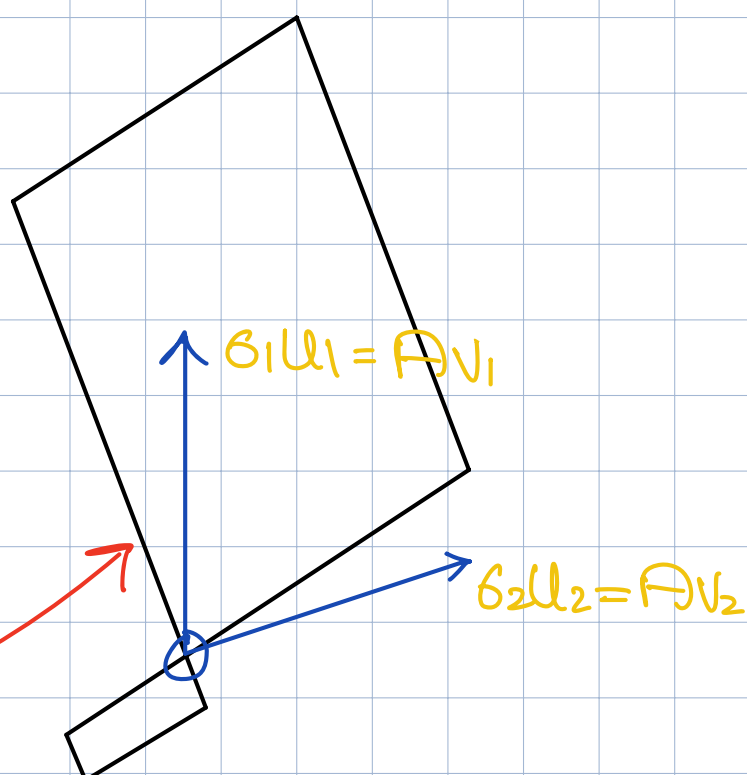


$$A = (\text{Orthogonal})(\text{Diagonal})(\text{Orthogonal})$$

Row-space $\mathbb{R}^n$



Col space $\mathbb{R}^m$



Orthogonal Basis
in Row A



Orthogonal Basis
in Col A

$$A \begin{bmatrix} v_1 & v_2 & \dots & v_r & \underbrace{v_{r+1} \dots v_m}_{\text{null space}} \end{bmatrix} = \begin{bmatrix} u_1 & u_2 & \dots & u_r & \dots & u_m \end{bmatrix} \begin{bmatrix} \sigma_1 & & & & & \\ & \sigma_2 & & & & \\ & & \ddots & & & \\ & & & \sigma_r & & \\ & & & & 0 & \end{bmatrix}$$

$$AV = U\Sigma$$

$$A = U\Sigma V^T$$

$$= U\Sigma V$$

$$A^T A = V \Sigma^T U^T U \Sigma V^T$$

$$= V \begin{bmatrix} \sigma_1^2 & & \\ & \sigma_2^2 & \\ & & \ddots \end{bmatrix} V^T$$

$\downarrow$ Eigen-vectors $\downarrow$ Eigen-values > 0

$$A A^T = U \Sigma V V^T \Sigma^T U^T$$

$$= U \begin{bmatrix} \sigma_1^2 & & \\ & \sigma_2^2 & \\ & & \ddots \end{bmatrix} U^T$$

$\downarrow$ Eigen-vectors $\downarrow$ Eigen-values > 0

Example,

$$A = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix}$$

v_1, v_2 orthonormal in Row A

u_1, u_2 orthonormal in Col A

$$\sigma_1, \sigma_2 > 0$$

$$Av_1 = \sigma_1 u_1$$

$$Av_2 = \sigma_2 u_2$$

$$A^T A = \begin{bmatrix} 25 & 7 \\ 7 & 25 \end{bmatrix}$$

$$A A^T = \begin{bmatrix} 32 & 0 \\ 0 & 18 \end{bmatrix}$$

$$\text{Eigen-vectors: } \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\text{Eigen-vectors: } \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\text{Eigen-values: } 32, 18$$

$$\text{Eigen-values} = 32, 18$$

$$\begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{32} & 0 \\ 0 & \sqrt{18} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

How is $\text{col}(A^T A) = \text{Row } A$?

$$v \in \text{col}(A^T A)$$

$$v = A^T A x$$

$$= A^T w$$

So, v in $\text{col } A^T A$ is formed rows of A

How is $\text{Row}(A A^T) = \text{Col } A$?

$$v \in \text{Row}(A A^T)$$

$$v = A A^T x$$

$$= A w$$

So, v in $\text{Row}(AA^T)$ formed from cols of A